

REQUEST FOR INFORMATION (RFI)

High-Speed Metal Stamping Press System for Brushless DC Motor Stator Lamination

For Tobyhanna Army Depot

1. INTRODUCTION

This Request for Information (RFI) is issued by the Tobyhanna Army Depot (TYAD) to solicit information and capabilities from potential sources for the procurement of a high-speed metal stamping press system. The system is intended for the lamination of brushless DC motor stators.

DISCLAIMER

THIS REQUEST FOR INFORMATION IS FOR INFORMATIONAL PURPOSES ONLY. THIS IS NOT A REQUEST FOR PROPOSAL (RFP) TO BE SUBMITTED. IT DOES NOT CONSTITUTE A SOLICITATION AND SHALL NOT BE CONSTRUED AS A COMMITMENT BY THE GOVERNMENT. RESPONSES IN ANY FORM ARE NOT OFFERS AND THE GOVERNMENT IS UNDER NO OBLIGATION TO AWARD A CONTRACT AS A RESULT OF THIS ANNOUNCEMENT. NO FUNDS ARE AVAILABLE TO PAY FOR PREPARATION OF RESPONSES TO THIS ANNOUNCEMENT. ANY INFORMATION SUBMITTED BY RESPONDENTS TO THIS TECHNICAL DESCRIPTION IS STRICTLY VOLUNTARY.

2. BACKGROUND

The Tobyhanna Army Depot has a potential requirement to procure a high-speed metal stamping press system configured to laminate brushless DC motor stators with a diameter of up to 50mm. The lamination process should utilize full-face bonding varnish (Backlack) for the lamination stacks bonding. Varnishes such as Thyssen Krupp Stabolit 70, Quaker Houghton QH Evertreat 7000, or similar varnish pre-applied to rolled non-oriented grain electrical steel sheet. Lamination system should include capability to self-feed electrical steel stock of thickness ranging from 0.2mm to 0.5mm thickness.

3. SCOPE OF INFORMATION REQUESTED

We are seeking information on commercially available systems that meet the following requirements. Please provide detailed responses to the following:

3.1. HIGH-SPEED PRESS SYSTEM

- **Press Specifications:** Provide detailed specifications for your high-speed metal stamping press system(s), including, but not limited to:
 - Tonnage capacity
 - Stroke rate (strokes per minute), including maximum and minimum speeds
 - Bed and slide dimensions.
 - Shut height adjustment range.
 - Deflection characteristics of the press bed and slide.
- **Drive System:** Describe the drive system of the press (e.g., conventional mechanical, servo-driven, link drive).

Explain the advantages of your proposed drive system for high-precision, high-speed lamination stamping.

- **Automation and Control:** Detail the automation and control features of the press system, including:
 - Programmable Logic Controller (PLC) and Human-Machine Interface (HMI).
 - Features for monitoring and adjusting the bottom dead center to compensate for thermal expansion.
 - Die protection and monitoring systems.
 - Systems for quick die changes.
- **Feed and Stacking Systems:** Provide information on the integrated or available systems for:
 - Coil feeding and material straightening.
 - Automated stacking of rotor and stator lamination.
 - Scrap removal.

3.2. LAMINATION AND BONDING PROCESS

- **Bonding Varnish Compatibility:** Describe your system's compatibility and experience with the applicable bonding varnishes.
- **Curing Process:** Detail the method for curing the bonding varnish, including temperature and pressure application, and cycle times.

Describe how your system ensures consistent and optimal bonding strength.

- **Lamination Quality:** Explain how your system ensures the quality of the laminated stator stacks, including:
 - Maintaining the magnetic properties of the electrical steel.
 - Minimizing eddy current losses.
 - Improving thermal conductivity and reducing noise and vibration.
- **Tooling:** Provide information on the types of dies (e.g., progressive, compound) suitable for this process and any recommendations for optimizing tool life.

3.3. SYSTEM INTEGRATION AND TURNKEY SOLUTIONS

- **Turnkey Systems:** Describe your capabilities in providing a complete turnkey solution, from coil handling to finished stator stack.
- **Integration:** If you provide individual components, describe how they can be integrated into a complete production line.
- **Support:** Detail the installation, training, and after-sales support provided.

4. VENDOR INFORMATION

Please provide the following information about your company:

- Company Name and Address.
- Point of Contact (Name, Title, Phone, and Email).
- A brief overview of your company's experience in providing similar high-speed stamping press systems.
- References from customers who have procured similar systems for electric motor manufacturing, if available.

SOURCING REQUIREMENTS

All materials, used in bearings, must not be sourced from prohibited foreign countries, including the People's Republic of China, the Russian Federation, the Islamic Republic of Iran, or the Democratic People's Republic of Korea.

5. SUBMISSION GUIDELINES

Interested businesses should submit a brief capabilities statement package (no more than ten 8.5 X 11-inch pages, font no smaller than 10 point) demonstrating ability to provide the products/systems listed in this Technical Description. Documentation should be in bullet format. No phone or email solicitations with regards to the status of the RFP will be accepted prior to its release.

Your response to this RFI, including any capabilities statement, shall be electronically submitted to the Lead Mechanical Engineer (Nicholas Paras) and TYAD UAS Project Manager (Christopher Volch) in either Microsoft Word or Portable Document Format (PDF), via email to corey.d.katsak.civ@army.mil & steven.wiltz.civ@army.mil no later than 12:00 p.m. Eastern Daylight Time (EDT) on 30 March 2026 and reference this synopsis number in subject line of e-mail and on all enclosed documents. Information and materials submitted in response to this request WILL NOT be returned. DO NOT SUBMIT CLASSIFIED MATERIAL.

If your organization has the potential capacity to provide the required products/systems, please provide the following information: 1) Organization name, address, primary points of contact (POCs) and their email address, FAX, Web site address, telephone number, and type of ownership for the organization; and 2) Tailored capability statements addressing the requirements of this notice, with appropriate documentation supporting claims of organizational and staff capability. If significant subcontracting or teaming is anticipated to deliver technical capability, organizations should address the administrative and management structure of such arrangements.

All data received in response to this RFI that is marked or designated as corporate or proprietary will be fully protected from any release outside the Government.